



COMPETENCY STANDARD

FOR

**ADVANCED GARMENTS PRODUCTION AND
LEAN MANAGEMENT**

(RMG SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

COPYRIGHT

The Competency Standards for Advanced Garments Production and Lean Management (RMG sector) is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment. Subsequently, they would be qualified and settled for a relevant job.

The document was developed under the Skills for Employment Investment Program (SEIP) and subsequently reviewed and updated with the engagement of occupation-specific industry experts to use in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)**. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

Public and private institutions may use the information contained in this standard for activities benefitting Bangladesh.

Other interested parties must obtain permission from the owner of this document for reproduction of information in any manner in whole or in part of this Skills Standard, in English or other languages.

This document is available at:

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance,
Probashi Kallyan Bhaban (Level-15 & 16)
71-72 Old Elephant Road, Eskaton Garden
Dhaka-1000
Phone: +8802 55138753-55, Fax: +88 02 55138752
Website: <https://sicip.gov.bd/>

INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (30 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-AGPLM-01-G	Apply Occupational Health and Safety (OHS) Practice in the Workplace	1 Identify OHS policies and procedures. 2 Apply personal health and safety practices. 3 Report hazards and risks. 4 Respond to emergencies	10
SICIP-RMG-AGPLM-02-G	Manage Personal and Professional Development	1 Interpret personal development skills. 2 Set and meet self-development priorities. 3 Maintain professional growth and development.	10
SICIP-RMG-AGPLM-03-G	Support Innovation and Manage Change	1 Identify needs for innovation in the workplace 2 Apply creative approach and solution 3 Support flexible and innovative ways of working 4 Adapt to emerging technological changes and opportunities	10
Total Hour			30 Hrs.

Sector Specific Competencies (10 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-AGPLM-01-S	Apply Green Practices in RMG Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hour			10 Hrs.

Occupation Specific Competencies (200 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-RMG-AGPLM-01-O	Identify Garment Construction	1. Interpret manufacturing process. 2. Identify industrial sewing machine and functions. 3. Identify stitches and seams. 4. Identify clothing materials	12
SICIP-RMG-AGPLM-02-O	Illustrate Garment Production Management	1. Define garment production 2. Interpret Key Aspects of Garments Production 3. Identify key performance indicators (KPIs) for production management	12
SICIP-RMG-AGPLM-03-O	Perform Work Study	1. Define work study. 2. Carry out method study in production floor. 3. Carry out work measurement in production floor. 4. Perform SMV calculation in production floor. 5. Perform production capacity and target calculation in production floor. 6. Perform efficiency calculation in production floor.	56
SICIP-RMG-AGPLM-04-O	Perform Line Layout	1. Carry out garment's operation breakdown. 2. Identify types of layouts 3. Practice skills matrix in production floor. 4. Carry out line layout.	20
SICIP-RMG-AGPLM-05-O	Implement Lean Management	1. Define lean management 2. Identify lean manufacturing waste 3. Apply lean tools 4. Interpret six sigma	48
SICIP-RMG-AGPLM-06-O	Apply Tools for Ensuring Quality	1. Identify tools of quality 2. Identify garment defects 3. Use tools for rectifying the defects	20
SICIP-RMG-AGPLM-07-O	Perform Production Planning and Optimization	1. Prepare TNA plan. 2. Perform plant capacity calculation. 3. Prepare inventory plan. 4. Prepare production schedule 5. Perform clothing material utilization. 6. Perform process optimization	32
Total Hours			200 Hrs.

COMPETENCY STANDARD: ADVANCED GARMENTS PRODUCTION AND LEAN MANAGEMENT

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-RMG-AGPLM-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are interpreted in the workplace including <u>personal protective equipment (PPE)</u> . 2.2 Common health issues are recognized. 2.3 Common safety issues are identified.
3. Report hazards and risks	3.1 Hazards and risks are identified. 3.2 Hazards and risks assessment and controls are interpreted
4. Respond to emergencies	4.1 Respond to alarms and warning devices. 4.2 <u>Emergency response plans and procedures</u> are responded to. 4.3 <u>First aid procedures</u> during emergency situations are identified.

Range of variables:

Variables	Range (may include but not limited to)
1. OHS policies	1.1 Organizational OHS policies 1.2 International OHS requirements 1.3 Fire safety rules and regulations

2. Emergency response plans and procedures	2.1 Firefighting procedures 2.2 Earthquake response procedures 2.3 Emergency response plans and procedures 2.4 Medical and first aid
3. First aid procedure	3.1 Washing of open wound 3.2 Washing chemically infected area 3.3 Applying bandage 3.4 Taking appropriate medicine
4. Personal protective equipment	4.1 Safety glasses 4.2 Ear plugs 4.3 Gloves 4.4 Apron 4.5 Helmet 4.6 Mask 4.7 Safety shoes

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	2.1 Identifying OHS policies and procedures 2.2 Applying personal health and safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Firefighting equipment 4.4 Emergency response manual

	4.5 First aid kits 4.6 Stationery 4.7 Learning manual.
--	--

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified OHS policies and procedures 1.2 applied personal health and safety practices 1.3 reported hazards and risks 1.4 responded to emergencies.
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MANAGE PERSONAL AND PROFESSIONAL DEVELOPMENT	Nominal Duration: 10 hrs.	Unit Code: SICIP-RMG-IELM-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to manage personal and professional development. It specifically includes the tasks of interpreting personal development skills, setting and meeting self-development priorities and maintaining professional growth and development.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret personal development skills	1.1 Objectives of personal development skills are described. 1.2 <u>Personal development skills</u> are identified. 1.3 Intra and Interpersonal relationships are maintained in the course of managing oneself. 1.4 Self-analysis is performed and personal development needs are identified.
2. Set and meet self-development priorities	2.1 Tasks are prioritized to achieve personal, team and organizational goals and objectives. 2.2 <u>Resources</u> are utilized efficiently and effectively to manage work priorities and commitments. 2.3 Economic usage and maintenance of facilities are followed as per established procedures.
3. Maintain professional growth and development	3.1 Pro activeness/zeal is demonstrated in fulfilling personal and professional growth requirements. 3.2 <u>Trainings and career opportunities</u> are identified and accessed based on job requirements. 3.3 <u>Recognitions</u> are sought/ received and demonstrated as proof of career advancement. 3.4 <u>Licenses and/or certifications</u> relevant to the job and career are obtained and renewed

Range of variables:

Variables	Range (may include but not limited to)
1. Personal Development skills	1.1 Problem-solving 1.2 Self-confidence 1.3 Adaptability 1.4 Integrity 1.5 Work ethic 1.6 Pro-activeness
2. Resources	2.1 Human 2.2 Financial

	2.3 Technology
3. Trainings and career opportunities	3.1 Participation in training programs 3.1.1 Technical 3.1.2 Supervisory 3.1.3 Managerial 3.1.4 Continuing Education 3.2 Serving as Resource Persons in conferences and workshops
4. Recognitions	4.1 Recommendations 4.2 Citations 4.3 Certificate of Appreciations 4.4 Commendations 4.5 Awards 4.6 Tangible and Intangible Rewards
5. Licenses and/or certifications	5.1 National Certificates 5.2 Certificate of Competency 5.3 Support Level Licenses 5.4 Professional Licenses

Curricular Content Guide

1. Underpinning knowledge	1.1 Importance of personal development skills. 1.2 Organizational policies relevant to training and professional growth. 1.3 Company operations, procedures and standards. 1.4 Resources in work environment.
2. Underpinning skill	2.1 Utilizing and improving personal development skills 2.2 Maintaining Intra and Interpersonal relationship 2.3 Utilizing communication skills 2.4 Prioritizing tasks in accordance with work commitment. 2.5 Utilizing resources efficiently and effectively
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Standard operating procedure 4.4 Workplace documents, signs and symbols 4.5 Codes of conduct

	4.6 Projector
	4.7 Stationery
	4.8 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 attained personal development skills. 1.2 maintained intra and interpersonal relationship in the course of managing oneself. 1.3 prioritized tasks according to work commitments. 1.4 identified training and career opportunities. 1.5 completed trainings based on the requirements of the industries. 1.6 acquired and maintained licenses and/or certifications according to the requirement of the qualification.
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: SUPPORT INNOVATION AND MANAGE CHANGE	Nominal Duration: 10 hrs.	Unit Code: SICIP-RMG-IELM-03-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to support innovation and manage change. It specifically includes the tasks of identifying needs for innovation in the workplace, applying creative approach and solution, supporting flexible and innovative ways of working and adapting to emerging changes and opportunities.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify needs for innovation in the workplace	1.1. The need for <u>innovation</u> in scope of work is recognized 1.2. The value of <u>innovative practices</u> in the workplace is recognized 1.3. Existing way of working in the workplace are interpreted. 1.4. Drawbacks of existing way of working are identified. 1.5. Benefits of change are identified to make it consistent with organizational requirements 1.6. Realistic timelines and targets for implementation of changes are set.
2. Apply creative approach and solution	2.1 Opportunities and creative approaches to implement innovation practices are identified 2.2 Creative approaches of coworkers pertaining to work practices are analyzed and incorporated 2.3 Innovation in accordance with organizational requirements changes are implemented
3. Support flexible and innovative ways of working	3.1 Maximum innovative opportunities are promoted. 3.2 Work assignments to facilitate innovative work skills are organized. 3.3 Team members are provided with guidance and coaching on innovation in the workplace. 3.4 Models of innovative work practice are provided and discussed 3.5 Appropriate environment for learning and innovation is maintained.
4. Adapt to emerging technological changes and opportunities	4.1 Usages of different <u>technologies</u> is determined based on job requirements. 4.2 Appropriate technology is selected as per work specifications. 4.3 Relevant technology is effectively used in carrying out functions. 4.4 Appropriate implementation tools are used as per task

	requirement.
--	--------------

Range of variables:

Variables	Range (may include but not limited to)
1. Innovative practices	1.1. Self-directed support to implement new ideas 1.2. Collaborative arrangement to apply new method of working 1.3. Making scope of work more efficient to use new technology
2. Innovation	2.1. New ideas 2.2. Different ideas 2.3. New methods of doing work 2.4. Use of new Technology
3. Technologies	3.1. Office technology 3.2. Industrial technology 3.3. System technology 3.4. Information technology 3.5. Training technology

Curricular Content Guide

1. Underpinning knowledge	1.1. Definition of innovation 1.2. Current practice in own scope of work 1.3. Workplace procedures 1.4. Support required to generate creative ideas 1.5. Difference between innovation and creativity 1.6. Common effects of change and innovation in the workplace 1.7. Industrial and organizational context of change 1.8. Organization's policies, plans, procedures and structure 1.9. Knowledge of resources required by the organization's operations 1.10. Awareness on technology and its function 1.11. Operating instructions 1.12. Applicable software 1.13. Company policy in relation to relevant technology 1.14. Technology adaptability
2. Underpinning skill	2.1. Identifying resources required for creativity and innovation 2.2. Contributing to brainstorming session 2.3. Identifying issues and concerns of one's scope of work 2.4. Encouraging co-workers to generate and develop ideas 2.5. Evaluating potential obstacles to and opportunities for creativity and innovation 2.6. Sharing of best practices related to innovation and creativity

	2.7. Applying relevant technology 2.8. Using update machine and Software 2.9. Acquiring troubleshooting skills Identifying issues 2.10. Identifying current industry standard diagnostic tools 2.11. Describing common malfunctions and resolutions. 2.12. Determining the root cause of a routine malfunction
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified need for innovation in the workplace 1.2 applied creative approach and solution 1.3 supported flexible and innovative ways of working 1.4 adapted to emerging technological changes and opportunities
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: APPLY GREEN PRACTICES IN RMG SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-RMG-IELM-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in RMG sector. It specifically includes the tasks of interpreting green concepts, minimizing resources in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices in RMG (Ready-Made Garment)</u> manufacturing are explained. 1.2 Key terms and symbols in RMG production processes are identified. 1.3 <u>Sources of environmental impacts</u> during RMG manufacturing activities are identified. 1.4 RMG manufacturing activities use are explained. 1.5 <u>Ways of mitigating environmental impacts</u> in RMG manufacturing are explained.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw material consumption in RMG manufacturing are documented. 2.2 <u>Recyclable and non-recyclable items</u> in RMG production processes are identified. 2.3 Procedures to reduce resource consumption in RMG manufacturing are implemented. 2.4 Sustainable alternatives in RMG manufacturing are explored and applied.
3. Implement waste management practices	3.1 <u>Different types of waste in RMG manufacturing</u> are identified. 3.2 Hazardous waste in RMG production is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Principles of green practices in RMG (Ready-Made Garment)	1.1 Reducing energy consumption in RMG manufacturing processes. 1.2 6R for waste management in RMG production: 1.2.1 Refuse unnecessary materials and processes.

	1.2.2 Reduce resource consumption and waste generation. 1.2.3 Reuse materials within the production process. 1.2.4 Recycle materials for future use in RMG manufacturing. 1.2.5 Recover usable resources from waste. 1.2.6 Repair damaged products or materials instead of discarding them. 1.3 Use of sustainable materials in RMG production with low environmental impact. 1.4 Recycling of materials used in the RMG manufacturing process. 1.5 Sustainable transportation methods for moving goods and materials within the RMG sector.
2. Sources of environmental impacts	2.1 Air Pollution in RMG Manufacturing 2.2 Water Pollution in RMG Manufacturing 2.3 Soil Degradation in RMG Manufacturing 2.4 Noise Pollution in RMG Manufacturing 2.5 Waste Generation in RMG Manufacturing 2.6 Resource Depletion in RMG Manufacturing
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment in RMG production processes. 3.2 Adopting renewable energy sources for RMG manufacturing operations. 3.3 Implementing site protection measures to prevent pollution and reduce environmental. 3.4 Using reusable materials in production processes. 3.5 Choosing sustainable materials with low environmental impact for garment production. 3.6 Using noise-reducing equipment and scheduling production work. 3.7 Optimizing logistics and delivery to reduce transportation emissions and improve overall supply chain efficiency.
4. Recyclable and non-recyclable items	4.1 Recyclable Items in RMG Manufacturing 4.1.1 Metal components. 4.1.2 Wooden materials. 4.1.3 Fabric scraps and textile fibers. 4.1.4 Plastic items, including buttons, tags, and packaging materials. 4.2 Non-Recyclable Items in RMG Manufacturing 4.2.1 Paints, coatings, and dyes. 4.2.2 Contaminated materials. 4.2.3 Treated wood and other materials.

5. Different types of waste in RMG manufacturing	5.1 Fabric waste from cutting and trimming processes. 5.2 Wood waste from packaging or display materials. 5.3 Metal waste from trims, buttons, zippers, and other hardware. 5.4 Textile waste, including scraps from garment production and defective products. 5.5 Plastic waste, such as packaging materials, labels, and plastic buttons. 5.6 Solid waste from chemical treatments or dyeing processes. 5.7 Packaging waste from raw materials and finished products.
--	--

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in RMG (Ready-Made Garment) manufacturing. 1.2 Sources of environmental impacts. 1.3 Ways of mitigating environmental impacts 1.4 Recyclable and non-recyclable items 1.5 Different types of waste in RMG manufacturing 1.6 Hazardous waste in RMG production 1.7 Green habits to reduce waste
2. Underpinning Skills	2.1 Explaining principles of green practices in RMG manufacturing 2.2 Identifying sources of environmental impacts 2.3 Explaining re-cyclable and non-recyclable items 2.4 Minimizing resources use in the workplace 2.5 Identifying different types of waste in RMG manufacturing 2.6 Disposing hazardous waste in RMG production 2.7 Practicing green habits to reduce waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimizing resources use in the workplace 1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: IDENTIFY GARMENT CONSTRUCTION	Nominal Duration: 12 Hrs.	Unit Code: SICIP-RMG-AGPLM-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to identify garment construction. It specifically includes the tasks of interpreting manufacturing process, identifying industrial sewing machines and functions, identifying stitches and seams, and identifying clothing materials.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret manufacturing process	1.1 <u>Fabric manufacturing process</u> is identified and explained. 1.2 <u>Garment manufacturing process</u> is identified and explained.
2. Identify industrial sewing machines and functions	2.1 <u>Types of industrial sewing machine</u> are identified. 2.2 Functions of industrial sewing machine are identified and described. 2.3 Types of <u>sewing needles</u> are identified. 2.4 Types of <u>attachments</u> used with industrial sewing machine are identified.
3. Identify stitches and seams	3.1 Types of <u>stitches</u> are identified as per garment style. 3.2 Types of <u>seams</u> are identified as per garment style. 3.3 Garment stitch quality is interpreted as per sample. 3.4 Garment seam quality is interpreted as per sample.
4. Identify clothing materials	4.1 Different types of <u>clothing materials</u> are identified. 4.2 <u>Bill of materials (BOM)</u> is identified. 4.3 Clothing materials are itemized and recorded as per the BOM (Bill of Material) sheet.

Range of Variables

Variable	Range (Includes but not limited to):
1. Fabric manufacturing process	1.1 Fiber 1.2 Yarn 1.3 Woven fabric 1.4 Knit fabric 1.5 Dying 1.6 Printing 1.7 Finishing 1.8 Yarn dyed fabrics

2. Garment manufacturing process	2.1 Design 2.2 Pattern making 2.3 Fit sample making 2.4 Production pattern making 2.5 Grading 2.6 Marker making 2.7 Fabric spreading 2.8 Fabric cutting 2.9 Cutting parts numbering and bundling 2.10 Sewing 2.11 Washing and embellishment 2.12 Finishing and packaging
3. Types of Industrial sewing machine	3.1 Single needle (chain and lock) 3.2 Double needle (chain and lock) 3.3 Over lock 3.4 Flat Lock 3.5 Feed of the arm 3.6 Multi needle chain stitch 3.7 Two step zigzags 3.8 Three step zigzags 3.9 Bar tack 3.10 Button hole 3.11 Button stitch 3.12 Eyelet
4. Sewing needles	4.1 DBx1 (Versatile for general stitching on medium-weight woven fabrics) 4.2 DBx5 (Designed for high-speed, heavy-duty straight stitching) 4.3 DCx1 (Used for parallel stitching on various fabric types) 4.4 DCx5 (Heavy-duty double-needle stitching for thick materials) 4.5 DCxK5 (For stretchy fabrics, like knits and lycra, using double-needle setup)
5. Attachments	5.1 Feed 5.2 Guide 5.3 Folder
6. Stitches	6.1 Chain 6.2 Lock 6.3 Hand
7. Seams	7.1 Super imposed 7.2 French 7.3 Lapped 7.4 Lap felled 7.5 Bound 7.6 Flat 7.7 Decorative 7.8 Edge neatening

8. Clothing materials	8.1 Fabric 8.2 Trims and accessories
9. Bill of materials (BOM)	9.1 Item Description 9.2 Material Specifications 9.3 Quantity Required 9.4 Unit of Measurement 9.5 Garment Parts 9.6 Trim and Accessories 9.7 Garment Size Information

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify and explain the fabric manufacturing process. 1.2 Identify and explain the garment manufacturing process. 1.3 Identify types of industrial sewing machines. 1.4 Identify and describe the functions of industrial sewing machines. 1.5 Identify different types of sewing needles. 1.6 Identify types of attachments used with industrial sewing machines. 1.7 Identify types of stitches as per garment style. 1.8 Identify types of seams as per garment style. 1.9 Interpret garment stitch quality as per sample. 1.10 Interpret garment seam quality as per sample. 1.11 Identify different types of clothing materials. 1.12 Identify the Bill of Materials (BOM).
2. Underpinning Skills	2.1 Performing fabric manufacturing process identification and explanation. 2.2 Performing garment manufacturing process identification and explanation. 2.3 Interpreting garment seam quality based on sample. 2.4 Identifying different types of clothing materials. 2.5 Identifying the Bill of Materials (BOM). 2.6 Itemizing and recording clothing materials as per the BOM.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual)

	4.2 Fabric (samples)
	4.3 Garment (samples)
	4.4 Bill of Materials (samples)

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted manufacturing process 1.2 identified industrial sewing machines and functions 1.3 identified stitches and seams 1.4 identified clothing materials
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by nationally certified assessor or occupation-specific industry expert.

Unit of Competency: ILLUSTRATE GARMENT PRODUCTION MANAGEMENT	Nominal Duration: 12 Hrs.	Unit Code: SICIP-RMG-AGPLM-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to illustrate garment production management. It specifically includes the tasks of defining garment production, interpreting key aspects of garments production, and identifying key performance indicators (KPIs) for production management.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Define garment production	1.1 Garment production is defined. 1.2 <u>Factors of production</u> are identified and understood. 1.3 Production process is explained. 1.4 Production efficiency and quality standards are understood.
2. Interpret Key Aspects of Garments Production	2.1 Key aspects of garment production are identified. 2.2 Importance of quality control in each stage of production is understood. 2.3 <u>Labor and material costs</u> in garment production are recognized to ensure cost efficiency. 2.4 Concept of lean manufacturing is identified. 2.5 Integration of machinery and technology into production processes is understood. 2.6 Production performance metrics are interpreted and.
3. Identify key performance indicators (KPIs) for production management	3.1 <u>Key Performance Indicators (KPI)</u> is defined. 3.2 Production output is identified and understood as a primary KPI for efficiency. 3.3 Quality control metrics are explained as key indicators of product quality. 3.4 Machine downtime is understood to identify areas for equipment maintenance and process improvement. 3.5 Labor productivity is interpreted to improve workforce efficiency.

Range of Variables

Variable	Range (Includes but not limited to):
1. Factors of production	1.1 Man power 1.2 Machinery 1.3 Facility 1.4 Materials

2. Labor and material costs	2.1 Direct Labor 2.2 Indirect Labor 2.3 Fixed cost 2.4 Overhead Costs 2.5 Transportation Costs 2.6 Import/Export Duties 2.7 Fabric Costs 2.8 Trim Costs 2.9 Accessories Costs
3. Key Performance Indicators (KPI)	3.1 Production Efficiency 3.2 Defects Rate 3.3 On-Time Delivery 3.4 Capacity Utilization 3.5 Labour Productivity 3.6 Inventory Turnover 3.7 Machine Downtime 3.8 Cost per Minute 3.9 Lead Time 3.10 Workforce Turnover Rate 3.11 Right first time 3.12 Cut to ship ratio 3.13 Merker efficiency

Curricular Content Guide

1. Underpinning Knowledge	1.1 Define garment production as the process of creating finished clothing products. 1.2 Identify and understand the factors of production. 1.3 Explain the production process from raw material to finished garment. 1.4 Understand production efficiency and quality standards for optimal output. 1.5 Identify key aspects of garment production. 1.6 Understand the importance of quality control at each production stage. 1.7 Recognize labor and material costs. 1.8 Identify the concept of lean manufacturing. 1.9 Interpret production performance metrics. 1.10 Identify production output as the primary KPI for measuring efficiency. 1.11 Explain quality control metrics as key indicators of product quality. 1.12 Understand machine downtime as an indicator for equipment maintenance and process optimization.
2. Underpinning Skills	2.1 Defining garment production processes and principles. 2.2 Understanding factors of production. 2.3 Explaining the production process in garment

	manufacturing. 2.4 Understanding production efficiency and quality standards. 2.5 Recognizing labor and material costs in garment production to ensure cost efficiency. 2.6 Identifying the concept of lean manufacturing. 2.7 Interpreting production performance metrics for operational analysis. 2.8 Defining Key Performance Indicators (KPIs) in garment production. 2.9 Understanding machine downtime to identify areas for equipment maintenance and process improvement.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Laptop/computer 4.3 Learning materials 4.4 KPI Template

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 defined garment production 1.2 interpreted key aspects of garments production 1.3 identified key performance indicators (KPIs) for production management
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM WORK STUDY	Nominal Duration: 56 Hrs.	Unit Code: SICIP-RMG-AGPLM-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform work study. It specifically includes the tasks of defining work study, carrying out method study in production floor, carrying out work measurement in production floor, performing SMV calculation in production floor, performing production capacity and target calculation in production floor and performing efficiency calculation in production floor.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Define work study	1.1 Work study is defined. 1.2 Importance of work study is explained. 1.3 Basic procedure of work study is identified and described.
2. Carry out method study in production floor	2.1 Method study is defined 2.2 Improving production efficiency and reducing waste is understood. 2.3 <u>Method study procedures</u> are followed. 2.4 Primary and secondary questioning techniques are applied to gather relevant data. 2.5 Data collection methods are utilized to assess current production methods. 2.6 Potential improvements to the production process are identified based on the findings from the method study. 2.7 Results of the method study are analyzed and documented.
3. Carry out work measurement in production floor	3.1 Purpose of work measurement is understood and explained. 3.2 <u>Work measurement techniques</u> are applied to measure task performance. 3.3 Work measurement procedure is followed. 3.4 Employee ratings and allowances for work content are accounted. 3.5 Data collected during work measurement is analyzed. 3.6 Recommended improvements based on work measurement findings are implemented.
4. Perform SMV calculation in production floor	4.1 <u>Tools for SMV calculation</u> are identified. 4.2 Procedure for SMV calculation is identified. 4.3 Cycle check data is identified and computed 4.4 <u>SMV calculation formula</u> is identified and interpreted. 4.5 SMV calculation is carried out according to formula.
5. Perform production	5.1 <u>Production capacity</u> on process, line and factory are

capacity and target calculation in production floor	identified. 5.2 Production capacity calculation formula is identified interpreted. 5.3 Production capacity is calculated as per formula. 5.4 <u>Production target calculation formula</u> is identified and interpreted. 5.5 Production target is calculated as per formula.
6. Perform efficiency calculation in production floor	6.1 <u>Efficiency calculation method</u> is identified. 6.2 Efficiency calculation formula is interpreted. 6.3 Efficiency calculations are carried out according to formula.

Range of Variables

Variable	Range (Includes but not limited to):
1. Method study procedures	1.1 Select 1.2 Record 1.3 Examine 1.4 Develop 1.5 Evaluate 1.6 Define 1.7 Install 1.8 Maintain
2. Work measurement techniques	2.1 Work sampling 2.2 Time study 2.3 Predetermined Motion Time Standards (PMTS) 2.4 Standard rate
3. Tools for SMV calculation	3.1 Stop watch 3.2 Paper 3.3 Pen 3.4 Pencils 3.5 Calculator 3.6 Computer
4. Standard Minute Value (SMV) calculation formula	4.1 Cycle time 4.2 Observed time 4.3 Basic time 4.4 Performance rating 4.5 Allowances
5. Production capacity	5.1 Working hour 5.2 SMV 5.3 Total SMV earners 5.4 Working days
6. Production target calculation formula	6.1 Production capacity 6.2 Efficiency 6.3 Absenteeism
7. Efficiency calculation method	7.1 Production output

	7.2 SMV
	7.3 SMV earners
	7.4 Working hours

Curricular Content Guide

1. Underpinning Knowledge	1.1 Define work study as the systematic analysis of work processes. 1.2 Explain how work study enhances efficiency, reduces costs, and eliminates waste. 1.3 Describe method study as a technique to improve work methods and increase efficiency. 1.4 Understand production efficiency by focusing on reducing waste and optimizing resources. 1.5 Understand work measurement by measuring the time required. 1.6 $SMV = (\text{Observed Time} + \text{Allowances})$ to determine the standard time for a task. 1.7 $\text{Efficiency} = (\text{Actual Output} / \text{Target Output}) \times 100$ to measure performance.
2. Underpinning Skills	2.1 Defining work study principles and procedures. 2.2 Defining method study and its relevance in production. 2.3 Following method study procedures accurately. 2.4 Applying primary and secondary questioning techniques to gather relevant data. 2.5 Utilizing data collection methods to assess current production methods. 2.6 Analyzing and documenting results from the method study. 2.7 Applying work measurement techniques to measure task performance. 2.8 Following the work measurement procedure correctly. 2.9 Analyzing data collected during work measurement. 2.10 Implementing recommended improvements based on work measurement findings. 2.11 Carrying out SMV calculation according to the formula. 2.12 Calculating production capacity as per the formula. 2.13 Calculating production target as per the formula. 2.14 Carrying out efficiency calculations according to the formula.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace

	3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Calculator 4.4 Projector 4.5 Stationery 4.6 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 defined work study 1.2 carried out method study in production floor. 1.3 carried out work measurement in production floor. 1.4 performed SMV calculation in production floor. 1.5 performed production capacity and target calculation in production floor. 1.6 performed efficiency calculation in production floor
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM LINE LAYOUT	Nominal Duration: 20 Hrs.	Unit Code: SICIP-RMG-AGPLM-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform line layout. It specifically includes the tasks of carrying out garment's operation breakdown, identifying types of layouts, practicing skills matrix in production floor and carrying out line layout.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out garment's operation breakdown	1.1 <u>Garments operation breakdown</u> is identified and explained as per styling. 1.2 Garments operation breakdown is performed as per production sample.
2. Identify types of layouts	2.1 Importance of layout in production line is explained. 2.2 <u>Types of layouts</u> are identified as per job requirements.
3. Practice skills matrix in production floor	3.1 Skills matrix is defined and explained. 3.2 Skills matrix is created and maintained. 3.3 Workers' <u>skill levels</u> are assessed and recorded in the skills matrix. 3.4 Skills matrix is used to assign tasks based on workers' competencies. 3.5 Regular updates are made to the skills matrix. 3.6 Skills matrix is reviewed periodically to ensure that it aligns with production requirements.
4. Carry out line layout	4.1 <u>Line layout</u> is interpreted as per styling. 4.2 Line balancing is interpreted and maintained. 4.3 Line layout is performed in accordance with the production sample.

Range of Variables

Variable	Range (Includes but not limited to):
1. Garments operation breakdown	1.1 Style of the garments 1.2 Front part 1.3 Back part 1.4 Assembling part 1.5 Make section
2. Types of layouts	2.1 Straight 2.2 Side by side 2.3 U- shaped 2.4 Modular 2.5 Progressive Bundle System (PBS)

	2.6 Unit Production System (UPS)
3. Skill levels	3.1 Sewing Machine Type 3.2 Process Type 3.3 Efficiency
4. Line layout	4.1 Operation breakdown 4.2 Machine selection 4.3 Operator selection 4.4 Standard Minute Value (SMV) 4.5 Skills matrix

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify and explain garments operation breakdown based on styling. 1.2 Explain the importance of layout in the production line. 1.3 Identify the types of layouts according to job requirements. 1.4 Define and explain the skills matrix. 1.5 Interpret line layout as per styling. 1.6 Interpret and maintain line balancing. 1.7 Line layout in accordance with the production sample.
2. Underpinning Skills	2.1 Performing garment operation breakdown as per production sample. 2.2 Defining and explaining the skills matrix. 2.3 Creating and maintaining the skills matrix. 2.4 Assessing and recording workers' skill levels in the skills matrix. 2.5 Using the skills matrix to assign tasks based on workers' competencies. 2.6 Making regular updates to the skills matrix. 2.7 Reviewing the skills matrix periodically to ensure alignment with production requirements. 2.8 Performing line layout in accordance with the production sample.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure (samples) 4.3 Projector

	4.4 Layouts
	4.5 Skills matrix template
	4.6 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 carried out garment's operation breakdown 1.2 identified types of layouts 1.3 practiced skills matrix in production floor. 1.4 carried out line layout.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: IMPLEMENT LEAN MANUFACTURING SYSTEM	Nominal Duration: 48 Hrs.	Unit Code: SICIP-RMG-AGPLM-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to implement lean manufacturing system. It specifically includes the tasks of defining lean management, identifying lean manufacturing waste, applying lean tools and interpreting six sigma.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Define lean management	1.1 <u>Basic lean management systems</u> are defined. 1.2 Lean management system is interpreted to increase efficiency. 1.3 Key principles of lean management are understood.
2. Identify lean manufacturing waste	2.1 <u>Types of waste</u> in lean manufacturing are identified. 2.2 Impact of each type of waste on production efficiency and cost is understood. 2.3 Lean manufacturing wastage solutions are identified.
3. Apply lean tools	3.1 Basic lean manufacturing <u>tools and techniques</u> are identified. 3.2 Impact of application of tools and techniques to lean manufacturing is described. 3.3 Lean manufacturing tools are applied to enhance organizational productivity and efficiency.
4. Interpret six sigma	4.1 Six sigma is identified and interpreted. 4.2 <u>DMAIC Framework</u> is identified and interpreted.

Range of Variables

Variable	Range (Includes but not limited to):
1. Basic lean management systems	1.1 Value 1.2 Value stream mapping 1.3 Flow 1.4 Pull 1.5 Perfection
2. Types of waste	2.1 Over production 2.2 Over processing 2.3 Excess transportation 2.4 Excess inventory 2.5 Excess motion 2.6 Waiting 2.7 Re- work 2.8 Unused talents
3. Tools and techniques	3.1 Value stream mapping (VSM)

	3.2 Workplace organization (5S) 3.3 Visual management 3.4 Kanban and super market 3.5 Standardization of work process 3.6 Root cause analysis 3.7 Cellular manufacturing 3.8 Quick changeover 3.9 Overall equipment effectiveness 3.10 Problem solving 3.11 TPM 3.12 Kaizen
4. DMAIC Framework	4.1 Define 4.2 Measure 4.3 Analyze 4.4 Improve 4.5 Control

Curricular Content Guide

1. Underpinning Knowledge	1.1 Define basic lean management systems. 1.2 Interpret lean management systems to increase efficiency. 1.3 Understand the key principles of lean management. 1.4 Identify types of waste in lean manufacturing. 1.5 Understand the impact of each type of waste on production efficiency and cost. 1.6 Identify lean manufacturing wastage solutions. 1.7 Identify basic lean manufacturing tools and techniques. 1.8 Describe the impact of applying lean tools and techniques to manufacturing. 1.9 Identify and interpret Six Sigma. 1.10 Identify and interpret the DMAIC framework.
2. Underpinning Skills	2.1 Defining basic lean management systems. 2.2 Interpreting lean management systems to increase efficiency. 2.3 Understanding the impact of each type of waste on production efficiency and cost. 2.4 Applying lean manufacturing tools to enhance organizational productivity and efficiency. 2.5 Identifying and interpreting Six Sigma principles. 2.6 Identifying and interpreting the DMAIC (Define, Measure, Analyze, Improve, Control) framework.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn

	3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Lean tools 4.4 DMAIC framework

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 defined lean management 1.2 identified lean manufacturing waste 1.3 applied lean tools 1.4 interpreted six sigma
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY TOOLS FOR ENSURING QUALITY	Nominal Duration: 20 Hrs.	Unit Code: SICIP-RMG-AGPLM-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply tools for ensuring quality. It specifically includes the tasks of identifying tools of quality, identifying garment defects and using tools for rectifying the defects.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify tools of quality	1.1 <u>Basic tools of quality</u> are identified according to the specific requirements. 1.2 Tools for quality control are selected based on the quality standards to be achieved. 1.3 The usage of tools is aligned with organizational policies.
2. Identify garment defects	2.1 <u>Garment defects</u> are inspected and identified according to quality standards. 2.2 Defects are recorded accurately, including the type and location of the defect. 2.3 Defective garments are segregated from non-defective garments to prevent further use.
3. Use tools for rectifying the defects	3.1 Defective garments are examined to determine the appropriate tool required for rectification. 3.2 The defect rectification process is conducted in a manner that minimizes disruption. 3.3 Rectified garments are inspected. 3.4 All rectification actions and tool usage are documented for future reference and quality control.

Range of Variables

Variable	Range (Includes but not limited to):
1. Basic tools of quality	1.1 Check sheet 1.2 Control chart 1.3 Histogram 1.4 Cause and effect diagram 1.5 Pareto chart 1.6 Scatter diagram 1.7 Flow chart
2. Garment defects	2.1 Fabric defects: 2.1.1 Drop stitches 2.1.2 Dye marks 2.1.3 Laddering 2.1.4 Stains

	2.1.5 Bad selvedge 2.1.6 Shade variation 2.2 Sewing defects: 2.2.1 Skipped Stitches 2.2.2 Broken Threads 2.2.3 Puckering 2.2.4 Needle Damage 2.2.5 Seam Slippage 2.2.6 Uneven Stitch 2.2.7 Fabric Shifting 2.2.8 Untrimmed thread 2.2.9 Thread Tension Issues 2.2.10 Double Stitching 2.3 Trim defects: 2.3.1 Trim broken 2.3.2 Trim differs 2.3.3 Trim bleeding
--	--

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify basic tools of quality according to specific requirements. 1.2 Ensure the usage of tools aligns with organizational policies. 1.3 Inspect and identify garment defects according to quality standards. 1.4 Segregate defective garments from non-defective ones to prevent further use. 1.5 Rectified garments to ensure quality standards are met. 1.6 Rectification actions and tool usage for future reference and quality control.
2. Underpinning Skills	2.1 Selecting tools for quality control based on quality standards to be achieved. 2.2 Aligning the usage of tools with organizational policies. 2.3 Inspecting and identifying garment defects according to quality standards. 2.4 Recording defects accurately, including type and location. 2.5 Segregating defective garments from non-defective garments to prevent further use. 2.6 Examining defective garments to determine the appropriate tool required for rectification. 2.7 Conducting the defect rectification process in a manner that minimizes disruption. 2.8 Inspecting rectified garments for quality assurance. 2.9 Documenting all rectification actions and tool usage for future reference and quality control.

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Basic tools of quality 4.3 Garments defects sample 4.4 Work sheets

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified tools of quality 1.2 identified garment defects 1.3 used tools for rectifying the defects
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM PRODUCTION PLANNING AND OPTIMIZATION	Nominal Duration: 32 Hrs.	Unit Code: SICIP-RMG-AGPLM-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform production planning and optimization. It specifically includes the tasks of preparing TNA plan, performing plant capacity calculation, preparing inventory plan, performing production schedule, performing clothing material utilization and performing process optimization.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare TNA plan	1.1 <u>Time and Action (TNA) plan</u> is interpreted based on lead time. 1.2 TNA plan for a selected order is prepared. 1.3 TNA plan is reviewed and adjusted as needed.
2. Perform plant capacity calculation	2.1 <u>Plant capacity calculation formula</u> is identified. 2.2 Plant capacity formula is interpreted. 2.3 Plant capacity calculation is carried out. 2.4 Results of the plant capacity calculation are reviewed.
3. Prepare inventory plan	3.1 Importance of inventory planning is explained. 3.2 <u>Types of inventories</u> are identified. 3.3 Material Requirement Planning (MRP) is described. 3.4 Inventory plan is prepared. 3.5 Stock levels are monitored and adjusted.
4. Prepare production schedule	4.1 <u>Production scheduling</u> is explained. 4.2 Production schedule is carried out. 4.3 Required production data is collected. 4.4 Production schedule is prepared as per production scheduling. 4.5 Production schedule is reviewed for accuracy. 4.6 Finalized production schedule is communicated to relevant departments for execution.
5. Apply ERP software in production planning	5.1 Functionality of the ERP system for production planning is understood. 5.2 Production data is input and managed in the ERP software. 5.3 ERP software is utilized for production scheduling. 5.4 Inventory levels are monitored and tracked in real-time using the ERP software. 5.5 Reports are generated using ERP software to track key production metrics. 5.6 Production data is analyzed using ERP-generated reports in the production process.
6. Perform clothing	6.1 Efficiency of <u>clothing material consumption</u> is

material utilization	identified. 6.2 Material utilization is assessed and monitored throughout the production process. 6.3 Material utilization percentage is calculated based on the Bill of Materials (BOM) sheet. 6.4 Overall efficiency of material utilization is reported to stakeholders.
7. Perform process optimization	7.1 <u>Bottleneck process</u> is identified by analyzing production flow. 7.2 <u>Line balancing techniques</u> are followed to ensure that minimizing downtime. 7.3 <u>Line balancing tools</u> are used as per the line layout across the production line. 7.4 <u>Balancing loss formula</u> is interpreted and applied to calculate. 7.5 Balancing loss of the line is calculated using the identified formula. 7.6 Process optimization techniques are continuously monitored and adjusted.

Range of Variables

Variable	Range (Includes but not limited to):
1. Time and action plan	1.1 Lead time 1.2 Combined execution 1.3 Cutting 1.4 Sewing 1.5 Finishing and packing 1.6 Shipment
2. Plant capacity calculation formula	2.1 Total number of machines 2.2 Total hours factory runs a day 2.3 Total number of workers
3. Types of inventories	3.1 Fabrics 3.2 Trims and accessories 3.3 Finished goods 3.4 Work in process 3.5 Machinery 3.6 Tools and equipment
4. Material requirement planning	4.1 Technical package 4.2 Styling 4.3 Materials type 4.4 Required materials per unit 4.5 Sourcing lead time
5. Production scheduling	5.1 Lead time 5.2 Working days 5.3 Holidays 5.4 Calendar days

	5.5 Risk factors
6. Clothing material consumption	6.1 Types of fabric 6.2 Styling of apparel 6.3 Fabric width and length 6.4 Marker efficiency 6.5 Shrinkage of fabric 6.6 Size ratio breaks up 6.7 Trims type 6.8 Accessories type
7. Bottleneck process	7.1 Production Line Constraints 7.2 Material Shortages 7.3 Machine Downtime 7.4 Labor Shortages or Skill Gaps 7.5 Quality Control Delays 7.6 Waiting Time Between Processes 7.7 Supplier Delays 7.8 Data Processing Bottleneck 7.9 Inefficient Workflow 7.10 Overburdened Workstations 7.11 Complex Decision-Making or Approvals 7.12 Packaging and Shipping Delays
8 Line balancing techniques	8.1 Cycle Time-Based Balancing 8.2 Takt Time Calculation 8.3 Theoretical Minimum Workstations 8.4 Task Grouping/Bundle Techniques 8.5 Multi-Skilled Operators 8.6 Flexible Workstations 8.7 Time and Motion Study Analysis 8.8 Kanban System
9 Line balancing tools	9.1 Time and Motion Study 9.2 Work Sampling 9.3 Cycle Time Measurement Tools 9.4 Sewing Line Simulation Software 9.5 Work Balance Charts 9.6 Standard Operating Procedure (SOP) Documentation 9.7 Takt Time Calculators 9.8 Productivity Monitoring Tools 9.9 Gantt Charts for Task Allocation
10 Balancing loss formula	10.1 Number of allocated machines 10.2 Number of calculated machines

Curricular Content Guide

1. Underpinning Knowledge	1.1 Time and Action (TNA) plan helps schedule tasks based on lead time. 1.2 Plant capacity calculation determines the maximum output a plant can handle. 1.3 Inventory planning is crucial for maintaining stock
---------------------------	--

	levels. 1.4 Production scheduling organizes tasks and resources. 1.5 ERP systems streamline production planning, inventory management, and data analysis. 1.6 Line balancing ensures tasks are evenly distributed across workstations. 1.7 Bottleneck analysis and process optimization are key. 1.8 Continuous monitoring and adjustment of production processes.
2. Underpinning Skills	2.1 Preparing TNA plan for a selected order. 2.2 Reviewing and adjusting the TNA plan as needed. 2.3 Carrying out plant capacity calculation. 2.4 Reviewing results of the plant capacity calculation. 2.5 Preparing an inventory plan. 2.6 Monitoring and adjusting stock levels. 2.7 Carrying out production scheduling. 2.8 Collecting required production data. 2.9 Preparing production schedule as per production planning. 2.10 Inputting and managing production data in ERP software. 2.11 Utilizing ERP software for production scheduling. 2.12 Generating reports using ERP software to track key production metrics. 2.13 Analyzing production data using ERP-generated reports. 2.14 Calculating material utilization percentage based on the Bill of Materials (BOM) sheet. 2.15 Following line balancing techniques to minimize downtime. 2.16 Using line balancing tools as per the line layout across the production line. 2.17 Calculating balancing loss using the identified formula.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 TNA Plan (samples) 4.4 ERP Software 4.5 Bill of materials (BOM) sheet

	4.6 Calculator
	4.7 Line balancing tool

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared TNA plan 1.2 performed plant capacity calculation 1.3 prepared inventory plan 1.4 prepared production schedule 1.5 performed clothing material utilization 1.6 performed process optimization
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Advanced Garments Production and Lean Management
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)
- OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
The following machines are required to set up a sewing line as per industry standard		
1.	Single Needle Machine	02
2.	Over Lock Machine	02
3.	Flat Lock Machine	01
4.	Double Needle Machine	01
5.	Feed of the Arm Machine	01
6.	Multi Needle Chain Stitch Machine	01
7.	Button Stitch	01
8.	Button Hole Machine	01
9.	Eyelet Machine	01
10.	Digital Video Camera	01
11.	GSM Cutter	01
12.	GSM Scale	01
13.	Measuring Tape	25
14.	Multimedia Projector with Screen	01
15.	Stop Watch	25
16.	Calculator	10
17.	Desktop Computer/Laptop	05
18.	Printer	01

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

- The institute shall ensure organizing an internship program for a minimum of Forty (40) hours over Five (5) days in the RMG industry to cover the following units of competency.
 1. Perform Work Study
 2. Perform Line Layout

The internship shall specifically focus on the following elements of competency to be practiced on the production floor:

- a) Conduct method study
- b) Conduct work measurement
- c) Perform SMV (Standard Minute Value) calculation
- d) Calculate production capacity and target
- e) Calculate production efficiency
- f) Apply and practice the skills matrix

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Advanced Garments Production and Lean Management the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.